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Rheology testing systems and methods for operating the same

Abstract

A rheology system includes a rheometer including a lower plate and an upper plate, a manipulator including an arm, a loading end effector, a cleaning end effector, and a controller communicatively coupled to the rheometer and the manipulator, the controller including a processor and a computer readable and executable instruction set, which when executed, causes the processor to direct the manipulator to couple the loading end effector to the arm, direct the manipulator engage a specimen with the loading end effector, direct the manipulator to position the specimen on the lower plate of the rheometer, direct the upper plate to engage the specimen between the upper plate and the lower plate, direct the manipulator to couple the cleaning end effector to the arm, and direct the manipulator to engage the lower plate with the cleaning end effector.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage Entry under 35 U.S.C. § 371 of International Patent Application No. PCT/US2020/047936, filed Aug. 26, 2020, which claims priority to U.S. Provisional Patent Application No. 62/904,373, filed on Sep. 23, 2019, the entire disclosures of both of which are hereby incorporated by reference.

FIELD

(1) The present disclosure generally relates to rheology testing systems, and more particularly, to systems and methods for cleaning rheology testing systems between testing cycles.

BACKGROUND

(2) Characterizing physical properties of materials is useful in analyzing and improving chemical

formulations employed in the production of the materials as well as in analyzing and improving processes of manufacturing the materials. Characterizing the physical properties of particular materials may also assist in developing materials suitable for specific applications.

(3) In the case of some materials, such as polymers, biopolymers, and the like, it may be desirable to characterize the physical properties of the material by measuring the deformation and/or flow of the material under the influence of an applied force (e.g., a tensile force, a shear force, etc.). Conventional rheology tests may generally involve applying a specific stress to a material and monitoring the resultant deformation or strain, and generally include applying forces to the material that result in plastic flow of the material.

SUMMARY

(4) Rheology tests generally result in the plastic deformation of the material during the test. At the conclusion of the test, residue of the material samples must be removed from the testing apparatus before subsequent tests can be performed with new material samples. However, cleaning the residue of the material samples is a difficult and time consuming process, which generally increases testing costs and increases testing cycle time. Accordingly, a need exists for improved systems and methods for testing material samples and cleaning a testing apparatus.

(5) In one embodiment, a rheology system includes a rheometer including a lower plate and an upper plate, a manipulator including an arm, a loading end effector, a cleaning end effector, and a controller communicatively coupled to the rheometer and the manipulator, the controller including a processor and a computer readable and executable instruction set, which when executed, causes the processor to direct the manipulator to couple the loading end effector to the arm, direct the manipulator to engage a specimen with the loading end effector, direct the manipulator to position the specimen on the lower plate of the rheometer, direct the upper plate to engage the specimen between the upper plate and the lower plate, direct the manipulator to couple the cleaning end effector to the arm, and direct the manipulator to engage the lower plate with the cleaning end effector.

(6) In another embodiment, a method for operating a rheometer includes coupling a loading end effector to an arm of a manipulator, engaging a specimen with the loading end effector, positioning the specimen on a lower plate of a rheometer with the loading end effector, engaging the specimen with an upper plate of the rheometer, coupling a cleaning end effector to the arm of the manipulator, and engaging the lower plate with the cleaning end effector.

(7) Additional features and advantages will be set forth in the detailed description which follows, and in part will be readily apparent to those skilled in the art from that description or recognized by practicing the embodiments as described herein, including the detailed description which follows, the claims, as well as the appended drawings.

(8) It is to be understood that both the foregoing general description and the following detailed description are merely exemplary, and are intended to provide an overview or framework to understanding the nature and character of the claims. The accompanying drawings are included to provide a further understanding, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments, and together with the description serve to explain principles and operation of the various embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 schematically depicts a perspective view of a rheology system including a rheometer and a manipulator, according to one or more embodiments described and depicted herein;

(2) FIG. 2 schematically depicts a perspective view of a loading end effector of the manipulator of FIG. 1, according to one or more embodiments described and depicted herein;

- (3) FIG. 3 schematically depicts an enlarged view of a film engagement member of the loading end effector of FIG. 2, according to one or more embodiments described and depicted herein;
- (4) FIG. 4 schematically depicts a perspective view of a cleaning end effector of the manipulator of FIG. 1, according to one or more embodiments described and depicted herein;
- (5) FIG. 5 schematically depicts a secondary cleaning member of the rheology system of FIG. 1, according to one or more embodiments described and depicted herein;
- (6) FIG. 6 schematically depicts a control diagram of the rheology system of FIG. 1, according to one or more embodiments described and depicted herein; and
- (7) FIG. 7 schematically depicts a perspective view of the cleaning end effector of FIG. 4 approaching the rheometer of FIG. 1, according to one or more embodiments described and depicted herein.

DETAILED DESCRIPTION

(8) Embodiments described herein are generally directed to rheology systems including a rheometer and a manipulator that can be selectively coupled to a loading end effector and a cleaning end effector. The manipulator may automatically load the rheometer with specimens with the loading end effector, and may also load films to a cutting assembly to prepare the specimens for use with the rheometer. Once the rheometer has completed a rheology test on the specimen, the manipulator engages a lower plate and/or an upper plate of the rheometer with the cleaning end effector to remove residue of the specimen from the rheometer. In this way, rheology systems according to the present disclosure may automatically load and clean rheometers, without requiring removal of the lower plate and/or the upper plate from the rheometer. By allowing for automatic loading and cleaning of the rheometer, manual cleaning of the lower plate and/or the upper plate of the rheometer may be reduced or eliminated, thereby reducing testing time and cost. Furthermore, by cleaning the rheometer with the cleaning end effector, the rheometer may be utilized without requiring disposable intermediate components positioned between the lower plate and the specimen being tested, such as liners, which may improve thermal control of the specimen. Moreover, because the lower plate and/or the upper plate of the rheometer do not need to be removed and replaced between testing cycles, it is not necessary to re-heat new plates between testing cycles, reducing testing times and allowing for improved thermal control of the lower plate and/or the upper plate. Furthermore, by physically engaging the lower plate and/or the upper plate with the cleaning end effector, higher viscosity specimens which may not easily be removed chemically by solvents can be removed from the lower plate and/or the upper plate. These and other embodiments of rheology systems are disclosed in greater detail herein with reference to the appended figures.

(9) Referring initially to FIG. 1, a rheology system **100** is schematically depicted. In the embodiment depicted in FIG. 1, the rheology system **100** generally includes a rheometer **110** and a manipulator **130**. In some embodiments, the rheology system **100** further includes a cutting assembly **180** that is structurally configured to cut specimens for use with the rheometer **110**, as described in greater detail herein. In the embodiment depicted in FIG. 1, the rheology system **100** is positioned on work surface **22** (e.g., a table), however, it should be understood that this is merely an example.

(10) The rheometer **110** generally includes a lower plate **112** and an upper plate **114** spaced apart from the lower plate **112**. In the embodiment depicted in FIG. 1, the upper plate **114** is positioned above and is movable toward and away from the lower plate **112**. In embodiments, a specimen **12** may be sandwiched between the upper plate **114** and the lower plate **112**. While in the embodiment depicted in FIG. 1, the upper plate **114** is described as being movable with respect to the lower plate **112**, it should be understood that this is merely an example and in some embodiments, the lower plate **112** may be moved upward to the upper plate **114**.

(11) In some embodiments, the rheometer **110** further includes an upper actuator **118** coupled to the upper plate **114**. The upper actuator **118** may be directly coupled to the upper plate **114**, or may be coupled to the upper plate **114** through one or more intermediate components. The upper actuator

118 is operable to move the upper plate **114**, for example in a vertical direction toward and away from the lower plate **112**, and/or may be operable to rotate the upper plate **114** about the vertical direction. For example, in some embodiments, the rheometer **110** may test a specimen **12** positioned between the upper plate **114** and the lower plate **112** by rotating the upper plate **114** and monitoring shear within the specimen **12**. While in the embodiment depicted in FIG. 1, the rheometer **110** includes a single upper actuator **118**, it should be understood that the rheometer **110** may include any suitable number of actuators to move the upper plate **114** in the vertical direction and/or about the vertical direction.

(12) In some embodiments, the rheometer **110** includes a heating element **116** in thermal communication with the lower plate **112** and/or the upper plate **114**. In operation, the heating element **116** may apply thermal energy to the lower plate **112** and/or the upper plate **114**, respectively, such that thermal energy may be transmitted from the upper plate **114** and/or the lower plate **112** to a specimen **12** positioned between the upper plate **114** and the lower plate **112**. For example, in some rheology tests, it is desirable to understand the performance of a specimen **12** at a given temperature, and the heating element **116** may apply thermal energy to the specimen **12** via the lower plate **112** and/or the upper plate **114** to maintain the specimen **12** at a configurable temperature.

(13) While in the embodiment depicted in FIG. 1 the rheometer **110** includes a single heating element **116**, it should be understood that the rheometer **110** may include any suitable number of heating elements. The rheometer **110** may include one or more sensors configured to detect one or more attributes of the rheometer **110** and/or specimens tested by the rheometer **110**, and may include, for example, one or more temperature sensors and one or more sensors for detecting stress and strain on a specimen **12** being tested by the rheometer **110**. In some embodiments, the rheometer **110** may include a commercially available rheometer, such as DHR-2 rheometer available from TA Instruments.

(14) In the embodiment depicted in FIG. 1, the rheology system **100** includes two rheometers **110**, **110'**. By including two rheometers **110**, **110'**, throughput (e.g., the amount of specimens **12** tested over a predetermined amount of time) of the rheology system **100** may be increased. While in the embodiment depicted in FIG. 1, the rheology system **100** includes two rheometers **110**, **110'**, it should be understood that embodiments described herein may include any suitable number of rheometers and may include a single rheometer. Moreover, in the embodiment depicted in FIG. 1, the rheometers **110**, **110'** are substantially the same, however, in some embodiments, the rheology system **100** may include different rheometers that are structurally configured to perform different rheology tests.

(15) Still referring to FIG. 1, the manipulator **130** generally includes an arm **132** movably coupled to a base **134**. In the embodiment depicted in FIG. 1, the manipulator **130** includes a robot, such as a six-axis robot, however, it should be understood that this is merely an example. In operation, the manipulator **130** generally loads and unloads the cutting assembly **180** and the rheometers **110**, **110'**, as described in greater detail herein.

(16) The cutting assembly **180** is structurally configured to cut films **10** into specimens **12** for testing with the rheometers **110**, **110'**. The cutting assembly **180** generally includes at least one cutting member **182** that may include a laser, a blade, a cutting wheel, or the like. In operation, one or more films **10** may be provided to the cutting assembly **180**, for example via the manipulator **130**, and the cutting assembly **180** may cut the one or more films **10** to form specimens suitable for testing with the rheometers **110**, **110'**. As one example, the cutting assembly **180** may cut a stack of one or more films **10** to a desired size and shape to form the specimens for testing with the rheometers **110**, **110'**. In some embodiments, the films **10** and the specimens **12** are formed from a polymer, however, it should be understood that the films **10** and the specimens **12** may be formed from any material suitable for rheology testing, for example and without limitation, as glass, rubber, gums, waxes, concrete, gels, muds, biological materials, or the like.

(17) Referring to FIGS 2 and 3, a loading end effector **160** and an enlarged view of a film engagement member **162** of the loading end effector **160** are schematically depicted, respectively. The loading end effector **160** generally includes an arm engagement member **164** that can be selectively coupled to and is selectively removable from the arm **132** (FIG. 1) of the manipulator **130**. In embodiments, the loading end effector **160** further includes one or more film engagement members **162** that are structurally configured to engage a specimen **12** (FIG. 1) and/or a film **10** (FIG. 1). In embodiments, the one or more film engagement members **162** may be spaced apart from one another such that the loading end effector **160** can engage different size films.

(18) In some embodiments, the one or more film engagement members **162** may include suction cups or the like that engage the film **10** (FIG. 1) and/or the specimen **12** (FIG. 1). For example, in the embodiment depicted in FIG. 2, the loading end effector **160** includes one or more air inlets **166** that are fluidly coupled to a pump **192** and the one or more film engagement members **162**. In some embodiments, the loading end effector **160** includes one or more passageways **168** extending between the one or more film engagement members **162** and the one or more air inlets **166**. The one or more passageways **168** may include tubes or the like extending between the one or more air inlets **166** and the one or more film engagement members **163** such that fluid, such as air, may travel between the one or more air inlets **166** and the one or more film engagement members **162**. The pump **192**, in embodiments, applies a vacuum pressure to the one or more film engagement members **162**, such that film and/or specimens may be selectively drawn against the one or more film engagement members **162**.

(19) Referring to FIG. 4, a perspective view of a cleaning end effector **140** is schematically depicted. In embodiments, the cleaning end effector **140** includes a cleaning edge **144** coupled to an arm engagement member **142**. The arm engagement member **142** is selectively coupled to and selectively removable from the arm **132** of the manipulator **130** to selectively couple the cleaning end effector **140** to the manipulator **130**. The cleaning edge **144**, in operation, engages the lower plate **112** (FIG. 1) and/or the upper plate **114** (FIG. 1) of the rheometer **110** (FIG. 1) to remove residue of a specimen, as described in greater detail herein. In some embodiments, the cleaning edge **144** is formed from a material that is softer than the lower plate **112** (FIG. 1) and/or the upper plate **114** (FIG. 1), such that the cleaning edge **144** does not scratch or damage the lower plate **112** and/or the upper plate **114**. In some embodiments, the cleaning edge **144** is formed from brass or the like.

(20) In some embodiments, the cleaning edge **144** is coupled to the arm engagement member **142** through a resilient member **146**. Through the resilient member **146**, the cleaning edge **144** may be movable with respect to the arm engagement member **142**. More particularly, as force is applied to the cleaning edge **144**, the resilient member **146** may deform, thereby allowing some freedom of movement of the cleaning edge **144** with respect to the arm engagement member **142**. In embodiments, the resilient member **146** may include a spring, such as a compression spring, a torsion spring, an extension spring or the like.

(21) Referring to FIG. 5, a secondary cleaning member **170** of the rheology system **100** is schematically depicted. The secondary cleaning member **170** generally includes a brush **172** rotatably mounted to a base **174**. In some embodiments, the brush **172** is coupled to an actuator **176**, such as a motor or the like that is structurally configured to rotate the brush **172**. The actuator **176** may include, for example and without limitation a direct current (DC) motor, and alternating current (AC) motor, a pneumatic actuator, a hydraulic actuator, or the like. The brush **172**, in embodiments, generally includes bristles that may contact the lower plate **112** (FIG. 1) and/or the upper plate **114** (FIG. 1) to remove residue. In some embodiments, the bristles of the brush **172** may be softer than the lower plate **112** (FIG. 1) and/or the upper plate **114** (FIG. 1). In this way, the brush **172** may contact the lower plate **112** and/or the upper plate **114** without scratching the lower plate **112** and/or the upper plate **114**. In some embodiments, the secondary cleaning member **170** is structurally configured to be selectively coupled to the manipulator **130** (FIG. 1). In some

embodiments, the bristles of the brush **172** may be formed of brass or the like.

(22) Referring to FIG. **6**, a control diagram for the rheology system **100** is schematically depicted according to embodiments described herein. The rheology system **100** generally includes a controller **190**. As illustrated, the controller **190** includes a processor **198**, input/output hardware **197**, a network interface hardware **199**, a data storage component **196**, and a memory component **194**. The memory component **194** may be configured as volatile and/or nonvolatile memory and as such, may include random access memory (including SRAM, DRAM, and/or other types of RAM), flash memory, secure digital (SD) memory, registers, compact discs (CD), digital versatile discs (DVD), and/or other types of non-transitory computer-readable mediums. Depending on the particular embodiment, these non-transitory computer-readable mediums may reside within the controller **190** and/or external to the controller **190**.

(23) The memory component **194** may store operating logic, analysis logic, and communication logic in the form of one or more computer readable and executable instruction sets. The analysis logic and the communication logic may each include a plurality of different pieces of logic, each of which may be embodied as a computer program, firmware, and/or hardware, as an example. A local interface is also included in the controller **190**, and may be implemented as a bus or other communication interface to facilitate communication among the components of the controller **190**.

(24) The processor **198** may include any processing component operable to receive and execute instructions (such as from a data storage component **196** and/or the memory component **194**). The input/output hardware **197** may include and/or be configured to interface with microphones, speakers, a display, and/or other hardware.

(25) The network interface hardware **199** may include and/or be configured for communicating with any wired or wireless networking hardware, including an antenna, a modem, LAN port, wireless fidelity (Wi-Fi) card, WiMax card, ZigBee card, Bluetooth chip, USB card, mobile communications hardware, and/or other hardware for communicating with other networks and/or devices. From this connection, communication may be facilitated between the controller **190** and other computing devices.

(26) It should be understood that while the components in FIG. **6** are illustrated as residing within the controller **190**, this is merely an example. In some embodiments, one or more of the components may reside external to the controller **190**. It should also be understood that, while the controller **190** is illustrated as a single device, this is also merely an example.

(27) In embodiments, the controller **190** is communicatively coupled to one or more components of the rheology system **100**. For example, in the embodiment depicted in FIG. **6**, the controller **190** is communicatively coupled to the rheometer **110** (and/or **110'**), the manipulator **130**, the pump **192**, and/or the secondary cleaning member **170**.

(28) Referring now to FIGS. **1**, **2**, **4**, **5**, **6**, and **7**, a perspective view of the rheology system **100** is depicted in operation. As noted above, the controller **190** is communicatively coupled to the rheometer **110**, the manipulator **130**, the pump **192**, and/or the secondary cleaning member **170**. In embodiments, the controller **190** directs the manipulator **130** to couple the loading end effector **160** to the arm **132** of the manipulator **130**. With the loading end effector **160** coupled to the arm **132** of the manipulator **130**, the manipulator **130** may engage and move the film **10**. For example, in embodiments in which the loading end effector **160** suction cup film engagement members **162**, the controller **190** may direct the pump **192** to apply vacuum pressure to the film engagement members **162** to engage the film **10**.

(29) The manipulator **130** may position the film **10** at least partially within the cutting assembly **180** with the loading end effector **160**, and controller **190** may direct the cutting assembly **180** to cut the film **10** to form one or more specimens **12** for testing with the rheometers **110**, **110'**. As noted above, the cutting assembly **180** may cut the film **10** to any suitable shape or size for testing with the rheometers **110**, **110'**. The controller **190** may then direct the manipulator **130** to remove the cut specimen or specimens **12** from the cutting assembly **180**.

(30) In the embodiment depicted in FIG. 1, specimens **12** may be temporarily stored in a storage rack **20** before being placed within one of the rheometers **110**, **110'**. For example, in some embodiments, the cycle time of the cutting assembly **180** to cut the specimens **12** may not necessarily align with the cycle time of the rheometers **110**, **110'**, and accordingly, it may be desirable to temporarily store specimens **12** cut by the cutting assembly **180** before testing the specimens **12** in the rheometers **110**, **110'**. In this example, the manipulator **130** may remove the cut specimen or specimens **12** from the cutting assembly **180** and deposit the specimen or specimens **12** on the storage rack **20**.

(31) The controller **190** directs the manipulator **130** to position a specimen **12** in one of the rheometers **110**, **110'**, for example, via the loading end effector **160**. As one example, the manipulator **130** may move a specimen **12** from the storage rack **20** to one of the rheometers **110**, **110'**. As another example, the manipulator **130** may move a specimen **12** directly from the cutting assembly **180** to one of the rheometers **110**, **110'**. In embodiments, the controller **190** directs the manipulator **130** to position the specimen **12** between the upper plate **114** and the lower plate **112** of one of the rheometers **110**, **110'**.

(32) With the specimen **12** positioned between the upper plate **114** and the lower plate **112**, the rheometer **110**, **110'**, the controller **190** may direct the rheometer **110**, **110'** to perform a rheology test on the specimen. For example, the rheometer **110**, **110'** may lower the upper plate **114** toward the lower plate **112**, engaging the specimen **12** between the upper plate **114** and the lower plate **112**. In some embodiments, the upper plate **114** may rotate, thereby applying shear force to deform the specimen **12**. In some embodiments and as outlined above, the specimen **12** may be heated to a desired temperature by the heating element **116** of the rheometer **110**, **110'**.

(33) Once the rheology test of the specimen **12** is completed, controller **190** may direct the rheometer **110**, **110'** to move the upper plate **114** away from the lower plate **112**, as depicted in FIG. 7. The controller **190** then directs the manipulator **130** to couple the cleaning end effector **140** to the arm **132** of the manipulator **130**.

(34) With the cleaning end effector **140** coupled to the arm **132** of the manipulator **130**, the controller **190** directs the manipulator **130** to engage the lower plate **112** and/or the upper plate **114** with the cleaning end effector **140**. More particularly, the manipulator **130** engages the lower plate **112** and/or the upper plate **114** with the cleaning edge **144** of the cleaning end effector **140** and “scrapes” residue of the specimen **12** from the lower plate **112** and/or the upper plate **114**. In some embodiments, the manipulator **130** may control the force applied to the lower plate **112** and/or the upper plate **114** through positional control of the arm **132**. For example, the position of the lower plate **112** and/or the upper plate **114** may be communicated to the manipulator **130** through the controller **190**, such that the manipulator **130** may move the cleaning end effector **140** to engage but not damage the lower plate **112** and/or the upper plate **114**. In some embodiments, the manipulator **130** and/or the rheometer **110**, **110'** may include one or more force sensors structurally configured to detect the amount of force applied to the lower plate **112** and/or the upper plate **114**. The one or more force sensors may detect the amount of force applied to the lower plate **112** and/or the upper plate **114** via the cleaning end effector **140**. In response to the detected force, the controller **190** may direct the manipulator **130** to move the cleaning end effector **140** such that engagement between the cleaning end effector **140** and the lower plate **112** and/or the upper plate **114** is within a configurable threshold that does not damage the lower plate **112** and/or the upper plate **114** or the one or more sensors of the rheometer **110**, **110'**.

(35) After engaging the lower plate **112** and/or the upper plate **114**, residue of the specimen **12** may adhere to the cleaning edge **144** of the cleaning end effector **140**. Accordingly, before cleaning the lower plate **112** and/or the upper plate **114** in a subsequent testing cycle, the manipulator **130** may engage the cleaning edge **144** with a surface or cleaning device to remove the residue from the cleaning edge **144**. In some embodiments, different cleaning end effectors **140** may be used each testing cycle, allowing for off-line cleaning of the cleaning edge **144** of the cleaning end effector

140.

(36) In embodiments, subsequent to engaging the lower plate **112** and/or the upper plate **114** with the cleaning end effector **140**, the controller **190** directs the secondary cleaning member **170** to engage the lower plate **112** and/or the upper plate **114**. The controller **190** may further direct the secondary cleaning member **170** to rotate the brush **172**, engaging the lower plate **112** and/or the upper plate **114** with the brush **172**. In this way, the secondary cleaning member **170** may remove any remaining residue of the specimen **12** from the rheometer **110**, **110'**.

(37) In some embodiments and as noted above, the secondary cleaning member **170** can be selectively coupled to and removed from the manipulator **130**. In these embodiments, the controller **190** directs the manipulator **130** to couple the secondary cleaning member **170** to the arm **132** of the manipulator **130**. In some embodiments and as noted above, the secondary cleaning member **170** is independent from the manipulator **130**, and may engage the lower plate **112** and/or the upper plate **114** without use of the manipulator **130**.

(38) It should now be understood that embodiments described herein are directed to rheology systems including a rheometer and a manipulator that can be selectively coupled to a loading end effector and a cleaning end effector. The manipulator may automatically load the rheometer with specimens with the loading end effector, and may also load films to a cutting assembly to prepare the specimens for use with the rheometer. Once the rheometer has completed a rheology test on the specimen, the manipulator engages a lower plate and/or an upper plate of the rheometer with the cleaning end effector to remove residue of the specimen from the rheometer. In this way, rheology systems according to the present disclosure may automatically load and clean rheometers, without requiring removal of the lower plate and/or the upper plate from the rheometer. By allowing for automatic loading and cleaning of the rheometer, manual cleaning of the lower plate and/or the upper plate of the rheometer may be reduced or eliminated, thereby reducing testing time and cost. Furthermore, by cleaning the rheometer with the cleaning end effector, the rheometer may be utilized without requiring disposable intermediate components positioned between the lower plate and/or the upper plate and the specimen being tested, such as liners, which may improve thermal control of the specimen. Moreover, because the lower plate and/or the upper plate of the rheometer do not need to be removed and replaced between testing cycles, it is not necessary to re-heat new plates between testing cycles, reducing testing times and allowing for improved thermal control of the plate. Furthermore, by physically engaging the plates with the cleaning end effector, higher viscosity specimens which may not easily be removed chemically by solvents can be removed from the plates.

(39) It is noted that recitations herein of a component of the present disclosure being “structurally configured” in a particular way, to embody a particular property, or to function in a particular manner, are structural recitations, as opposed to recitations of intended use. More specifically, the references herein to the manner in which a component is “structurally configured” denotes an existing physical condition of the component and, as such, is to be taken as a definite recitation of the structural characteristics of the component.

(40) It is noted that terms like “preferably,” “commonly,” and “typically,” when utilized herein, are not utilized to limit the scope of the claimed invention or to imply that certain features are critical, essential, or even important to the structure or function of the claimed invention. Rather, these terms are merely intended to identify particular aspects of an embodiment of the present disclosure or to emphasize alternative or additional features that may or may not be utilized in a particular embodiment of the present disclosure.

(41) For the purposes of describing and defining the present invention it is noted that the terms “substantially” and “about” are utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. The terms “substantially” and “about” are also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the

basic function of the subject matter at issue.

(42) Having described the subject matter of the present disclosure in detail and by reference to specific embodiments thereof, it is noted that the various details disclosed herein should not be taken to imply that these details relate to elements that are essential components of the various embodiments described herein, even in cases where a particular element is illustrated in each of the drawings that accompany the present description. Further, it will be apparent that modifications and variations are possible without departing from the scope of the present disclosure, including, but not limited to, embodiments defined in the appended claims. More specifically, although some aspects of the present disclosure are identified herein as preferred or particularly advantageous, it is contemplated that the present disclosure is not necessarily limited to these aspects.

(43) It is noted that one or more of the following claims utilize the term “wherein” as a transitional phrase. For the purposes of defining the present invention, it is noted that this term is introduced in the claims as an open-ended transitional phrase that is used to introduce a recitation of a series of characteristics of the structure and should be interpreted in like manner as the more commonly used open-ended preamble term “comprising.”

Claims

1. A rheology system comprising: a rheometer comprising a lower plate and an upper plate; a manipulator comprising an arm; a loading end effector, wherein: the loading end effector comprises an arm engagement member coupled to one or more film engagement members; the arm engagement member comprises a member that can be selectively coupled to and removed from the arm; and the one or more film engagement members comprise suction cups fluidly coupled to a pump and configured such that vacuum pressure from the pump causes the specimen to be drawn against the one or more film engagement members; a cleaning end effector; and a controller communicatively coupled to the rheometer and the manipulator, the controller comprising a processor and a computer readable and executable instruction set, which when executed, causes the processor to: direct the manipulator to couple the loading end effector to the arm; direct the manipulator to engage a specimen with the loading end effector; direct the manipulator to position the specimen on the lower plate of the rheometer; direct the upper plate to engage the specimen between the upper plate and the lower plate; direct the manipulator to couple the cleaning end effector to the arm; and direct the manipulator to engage the lower plate with the cleaning end effector.

2. The system of claim 1, wherein the computer readable and executable instruction set, when executed, further causes the processor to direct the manipulator to engage the upper plate with the cleaning end effector.

3. The system of claim 2, wherein the processor directs the manipulator to engage the upper plate with the cleaning end effector while the upper plate is installed in the rheometer.

4. The system of either of claim 1, further comprising a cutting assembly communicatively coupled to the controller.

5. The system of claim 4, wherein the computer readable and executable instruction set, when executed, further causes the processor to: direct the manipulator to position a film at least partially within the cutting assembly with the loading end effector; direct the cutting assembly to cut the film to form the specimen; and direct the manipulator to remove the specimen from the cutting assembly.

6. The system of claim 1, wherein the cleaning end effector comprises an arm engagement member selectively engaged with the manipulator and a cleaning edge coupled to the arm engagement member.

7. The system of claim 6, wherein the cleaning edge is coupled to the arm engagement member through a resilient member.

8. The system of claim 6, wherein the cleaning edge comprises a material softer than the lower plate.

9. The system of claim 1, further comprising a secondary cleaning member communicatively coupled to the controller, and wherein the computer readable and executable instruction set, when executed, further causes the processor to: subsequent to directing the manipulator to engage the lower plate with the cleaning end effector, direct the secondary cleaning member to engage the lower plate.

10. The system of claim 9, wherein the computer readable and executable instruction set, when executed, causes the processor to direct the secondary cleaning member to engage the lower plate by directing the secondary cleaning member to rotate a brush that engages the lower plate.

11. The system of claim 9, wherein the computer readable and executable instruction set, when executed, further causes the processor to: prior to directing the secondary cleaning member to engage the lower plate, direct the manipulator to couple the secondary cleaning member to the arm of the manipulator.

12. The system of claim 1, wherein the processor directs the manipulator to engage the lower plate with the cleaning end effector while the lower plate is installed in the rheometer.

13. A method for operating a rheometer, the method comprising: coupling a loading end effector to an arm of a manipulator, wherein: the loading end effector comprises an arm engagement member coupled to one or more film engagement members; the arm engagement member comprises a member that can be selectively coupled to and removed from the arm; and the one or more film engagement members comprise suction cups fluidly coupled to a pump and configured such that vacuum pressure from the pump causes the specimen to be drawn against the one or more film engagement members; engaging a specimen with the loading end effector; positioning the specimen on a lower plate of a rheometer with the loading end effector; engaging the specimen with an upper plate of the rheometer; coupling a cleaning end effector to the arm of the manipulator; and engaging the lower plate with the cleaning end effector.

14. The method of claim 13, further comprising cutting a film with a cutting assembly to form the specimen, and removing the specimen from the cutting assembly with the loading end effector.

15. The method of claim 13, wherein the cleaning end effector comprises an arm engagement member selectively engaged with the manipulator and a cleaning edge coupled to the arm engagement member, and wherein engaging the lower plate comprises engaging the lower plate with the cleaning edge.

16. The method of claim 13, further comprising, subsequent to engaging the lower plate with the cleaning end effector, engaging the lower plate with a secondary cleaning member.

17. The method of claim 16, wherein engaging the lower plate with the secondary cleaning member comprises engaging the lower plate with a rotating brush of the secondary cleaning member.

18. The method of claim 16, further comprising prior to engaging the lower plate with the secondary cleaning member, coupling the secondary cleaning member to the manipulator.

19. The method of claim 13, wherein the cleaning end effector engages the lower plate while the lower plate is installed in the rheometer.
